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16 May 1955
SUPERSEDING
PA-PD-265
21 March 1953

MILITARY SPECIFICATION

POWDERS, METAL, ATOMIZED
(FOR USE IN AMMUNITION)

1. SCOPE

1.1 Scope.-This specification covers atomized metal powders for use in pyrotechnic compositions.

1.2 Classification.-The atomized metal powders shall be of the following types as specified (see 6.1):

Type I - Magnesium Powder, Atomized
Type II - Aluminum Powder, Atomized
Type III- Magnesium-Aluminum Alloy Powder 65/35, Atomized

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieves; Standard, for Testing Purposes.

MILITARY

MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

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(Copies of specifications, standards, drawings, and publications, required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

INTERSTATE COMMERCE COMMISSION

49 CFR 71-78 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition-Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71-78 (Rev. 1950).")

3. REQUIREMENTS

3.1 Type I, magnesium powder, atomized.

3.1.1 Volatile material at 105° Centigrade (C.).-The percentage of volatile material shall be 0.1 maximum (max.).

3.1.2 Carbides.-The percentage of carbides shall be 0.004 max..

3.1.3 Magnetic iron.-The percentage of magnetic iron shall be 0.05 max..

3.1.4 Alloy iron as Fe.-The percentage of alloyed iron as Fe shall be 0.05 max.

3.1.5 Free metallic magnesium.-The percentage of free metallic magnesium shall be 98.0 minimum (min.).

3.1.6 Zinc.-The percentage of zinc shall be 1.5 max.

3.1.7 Other impurities.-The percentage of other impurities shall be 0.3 max..

3.1.8 Granulation.-The granulation for type I, atomized magnesium powder shall be as specified in table I.

Table I

Nominal Mesh Size:	20-50		30-50		50-100		100-200		200-325	
Through U. S. Standard Sieve No.	Percent Min.	Max.	Percent Min.	Max.	Percent Min.	Max.	Percent Min.	Max.	Percent Min.	Max.
16	100	—								
20	95	—	95	—						
30	70	—	85	—	95	—	100	—		
50	—	20	—	40	70	—	90	—		
60	—	5	—	25						
80										
100			—	5	40	—	50	—		
140					—	20				
200					5	—	—	30	99.5	—
325							—	5	60	—

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3.2 Type II, aluminum powder, atomized.

3.2.1 Volatile material at 105°C.-The percentage of volatile material shall be 0.1 max.

3.2.2 Oil and grease.-The percentage of oil and grease shall be 0.2 max.

3.2.3 Zinc.-The percentage of zinc shall be 0.3 max.

3.2.4 Silicon.-The percentage of silicon shall be 0.25 max.

3.2.5 Free metallic aluminum.-The percentage of free metallic aluminum shall be 98.75 min.

3.2.6 Iron.-The percentage of iron shall be 0.5 max.

3.2.7 Aluminum oxide.-The percentage of aluminum oxide shall be 0.5 max.

3.2.8 Granulation.-Granulation shall be as specified in table II.

Table II

Nominal mesh size	200 - 325	
Through U.S. Standard Sieve No.	Percent Min.	Max.
200	99.5	—
325	60	—

3.3 Type III, magnesium-aluminum alloy powder, 65/35, atomized.

3.3.1 Aluminum.-The percentage of aluminum shall be 35 ± 5.

3.3.2 Magnesium.-The percentage of magnesium shall be 65 ± 5.

3.3.3 Total aluminum and magnesium.-The percentage of total aluminum and magnesium shall be 98.0 min.

3.3.4 Oil and grease.-The percentage of oil and grease shall be 0.2 max.

3.3.5 Silicon.-The percentage of silicon shall be 0.2 max.

3.3.6 Iron.-The percentage of iron shall be 0.1 max.

3.3.7 Granulation.-The granulation shall be as specified in table III.

Table III

Nominal mesh size	100 - 200		200 - 235	
Through U.S. Standard Sieve No.	Percent Min.	Max.	Percent Min.	Max.
30	100	—		
50	90	—		
100	50	—		
200	—	30	99.5	—
325	—	5	60	—

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3.3.8 Volatile material at 105°C.-The percentage of volatile material at 105°C. shall be 0.1 percent max.

3.4 Average particle diameter.-The average particle diameter shall be determined by the Fisher Sub-Sieve Sizer or the Picatinny Particle Sizer and shall be as specified in table IV. For nominal mesh size 200 - 325 the Fisher Sub-Sieve Sizer shall be used. For other nominal mesh sizes the Picatinny Particle Sizer shall be used.

Table IV

Nominal mesh size	20-50	30-50	50-100	100-200	200-325
Powder type	Average particle size, in microns				
Type I	450 ± 50	350 ± 50	200 ± 25	125 ± 25	22 ± 8
Type II	—	—	—	—	22 ± 8
Type III	—	—	—	125 ± 25	22 ± 8

3.5 Apparent density.-Atomized metal powders shall have a minimum apparent density of 1.0 gram (gm.) per milliliter (ml.).

3.6 Workmanship.- Atomized metal powders shall be manufactured by an atomizing process. The shape of type I, and type III particles shall be spheroidal such as is normally obtained by an atomizing process (see note 1). The shape of type II particles may be spheroidal or granular.

(Note 1.-"Atomization" is defined as the dispersion of a molten metal into a spray, the droplets of which are then allowed to freeze under the primary influence of surface tension. "Granular" is defined as particles that have approximately equi-dimensional non-spherical shapes.)

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot.-A lot shall consist of material obtained from a single batch of a batch process or from a single continuous process and shall be produced using a single set of operations and a single set of operating conditions.

4.2 Sampling.-Ten percent, but in no case more than 10 nor fewer than 3, of the containers shall be selected by the inspector so as to be representative of the lot. If there are fewer than 3 containers in the lot, all the containers shall be sampled. The material shall be mixed to a uniform consistency throughout and approximately 8 ounces of it shall be taken from each selected container. Each of these primary samples shall be placed in an airtight container and labeled so that the container from which it was taken can be identified. A composite sample of approximately 8 ounces shall be made from equal portions of the primary samples. The composite sample shall be thoroughly mixed and placed in an airtight container labeled to show the name of the material, manufacturer,

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plant, contract or purchase order number, lot number, and lot size. All acceptance tests shall be made on the composite sample. However, if it becomes apparent during sampling that the lot is not uniform, the inspector may require that any primary sample be tested for compliance with the requirements of the specification(see 6.2). All primary samples shall be held for future examination should the composite sample fail to comply with the requirements.

4.3 Inspection.-Inspection shall be as specified in Specification MIL-G-2550.

4.4 Tests applicable to type I, magnesium, powder, atomized.

4.4.1 Volatile material at 105°C.-Approximately 5,000 gm. of the sample shall be weighed into a tared dish and heated in an oven for 3 hours at 105°C, then cooled and weighed. The loss in weight shall be calculated as precent material volatile at 105°C.

4.4.2 Carbides.

4.4.2.1 Apparatus.-Apparatus similar to that shown on figure 1 shall be used.

4.4.2.2 Reagent.-A 0.25 gm. portion of gelatin shall be dissolved in hot water. The solution shall be diluted to 500 ml. and 500 ml. of 95-percent ethyl alcohol and 1.25 gm. hydroxylamine hydrochloride added.

4.4.2.3 Procedure.-A weighed portion of 50 gm. of magnesium powder shall be added to the reaction flask (A). Approximately 50 gm. of cuprous chloride shall be transferred to test tube (B). Nitrogen shall be passed through the apparatus to remove the air and shall be continued throughout the determination. The test tube shall be lowered and 15 ml. of concentrated ammonium hydroxide (NH_4OH) and 30 ml. of hydroxylamine hydrochloride reagent added to the reaction flask. The test tube shall be raised, the stopper inserted, 100 ml. of distilled water from dropping funnel (C) added and the heater switched on. The mixture shall be heated in the reaction flask just to boiling. A red or pink coloration of the cuprous ammonium chloride solution in the test tube indicates that the magnesium sample contains more than 0.004 percent carbides. Should the cuprous ammonium chloride solution become blue, due to the introduction of air, it shall be decolorized with successive 5 ml. portions of hydroxylamine hydrochloride reagent until the blue color disappears.

4.4.3 Magnetic iron.-Using 100 gm. of sample, a magnet or an electromagnet shall be passed through the powder until no more particles are taken up. The remainder of the sample shall be saved for the determination of alloyed iron. The particles adhering to the magnet shall be brushed onto a sheet of glazed paper and the magnetic separated from the nonmagnetic by passing the magnet under the paper and drawing the magnetic particles aside. This procedure shall be repeated to eliminate as much as possible any adhering particles of magnesium. The magnetic particles shall be transferred onto a balance pan, weighed and reported as metallic iron. In the event that this weight exceeds the requirements for the type of powder being tested, it shall be analyzed for iron as follows: Transfer the weighed material to a 250-ml. beaker and add 100 ml. of 5-percent sulfuric acid (H_2SO_4). Boil until dissolved. Cool. Pass the solution through a Jones reductor, taking the usual precautions to activate the amalgamated zinc. Wash with 50 ml. of 5-percent H_2SO_4 and 150 ml. of water. Titrate with standard 0.02 normal (N) potassium permanganate ($KMnO_4$). Calculate the percent metallic iron, as follows:

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$$\text{Percent metallic iron} = \frac{5.58\ VN}{W}$$

where:
V = ml. $KMnO_4$ solution used.
N = normality of $KMnO_4$ solution.
W = gm. of sample.

4.4.4 Alloyed iron as ferric oxide (Fe_2O_3).-An accurately weighed portion of approximately 10 gm. of the sample free from magnetic iron (see 4.4.3) shall be transferred to a 500-ml. flask. About 200 ml. of 1:5 H_2SO_4 shall be added slowly, to minimize effervescence and prevent spattering. The solution shall be allowed to stand until action ceases and all the magnesium has dissolved. The solution shall be titrated immediately with 0.02N $KMnO_4$ solution. The percent alloyed iron shall be calculated as follows:

$$\text{Percent alloyed iron (as } Fe_2O_3) = \frac{7.98\ VN}{W}$$

where:
V = ml. $KMnO_4$ required for sample.
N = normality of $KMnO_4$ solution.
W = gm. of sample.

4.4.5 Free metallic magnesium (eudiometer method).-An apparatus similar to that shown on figure 2 shall be assembled. The gas measuring buret shall be graduated in 0.2 ml. The reaction flask shall contain 100 ml. of 3-percent (by volume) of concentrated H_2SO_4 which has been saturated with hydrogen gas. Means shall be provided for dropping the sample into the acid. The weight of the sample to be used shall depend on the metal content and the temperature and pressure at which hydrogen is measured. (At standard conditions of temperature and pressure a 0.47-gm. sample of pure magnesium will yield about 470 ml. of hydrogen.) The weight of sample shall be roughly computed to give a volume of hydrogen within the capacity of the graduated portion of the gas buret. The system shall be in equilibrium with respect to temperature and pressure and shall be checked for any gas leakage before the determination is started. The weighed sample shall be folded in a piece of paper and placed in position. After comparing the reading of the thermometer in the reaction flask with a thermometer located in the vicinity of the apparatus the temperature shall be recorded. The volume shall also be recorded after the pressure is equalized by means of the leveling bulb. The confining liquid in the pipet is water saturated with hydrogen gas and containing 0.1-percent H_2SO_4 by volume and a few drops of methyl orange to facilitate the observation of the solution in the apparatus. The sample shall be allowed to drop into the acid and shall then be shaken. After the reaction has ceased, the system shall again be brought to equilibrium by cooling all parts of the system to the initial temperature within 1°C. and again equalizing the pressures, recording the volume of hydrogen gas evolved. The free metallic magnesium shall be calculated to percent as follows:

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$$\text{Percent free metallic magnesium} = \frac{\frac{R(P-Pw)}{760} \times \frac{273}{273 + T} \times 0.001086 \times 100}{W}$$

$$= \frac{0.03901R\ (P-Pw)}{W\ (273 + T)}$$

where:

R = reading of gas buret, in ml.
P = reading of barometer, in millimeters (mm.) of mercury(Hg).
Pw = vapor pressure of water at temperature T, in mm of Hg.
T = temperature of the system at equilibrium, in C^{o}.
W = gm. of sample.

4.4.6 Zinc.

4.4.6.1 Preparation for test.-Two gm. of the sample shall be weighed into a 400-ml. beaker and covered with water. Twenty ml. of concentrated hydrochloric acid (HCl) shall be added slowly. When the reaction has subsided, 1 ml. of concentrated nitric acid (HNO_3) shall be added and boiled until all copper is dissolved, then diluted to 200 ml. with distilled water, cooled to approximately 60°C. and saturated for 5 minutes with hydrogen sulfide (H_2S). The solution shall be filtered with a small amount of paper pulp and washed with hydrogen sulfide water containing a little HCl. The filtrate shall be boilded to expel the H_2S, cooled slightly, 10 gm. of citric acid added, diluted to 300 ml. with distilled water, and made neutral with ammonia, using methyl red as an indicator. Twenty-five ml. of a standard formic acid mixture (see note 2) shall be added the solution heated nearly to the boiling point and saturated with a rapid stream of H_2S for 15 minutes. After settling for 2 or 3 hours, the solution shall be filtered through a number (No.) 42 Whatman or equivalent, filter paper with a little ashless paper pulp and washed with hot diluted formic acid mixture (see note 3) which has been saturated with H_2S. The zinc content shall be determined by the gravimetric method using the zinc sulfide retained on the filter paper.

4.4.6.2 Procedure.-The filter paper containing the zinc sulfide shall be ignited in a tared ignited porcelain crucible at 700°C., cooled in a desiccator and weighed as zinc oxide. The percent zinc shall be calculated as follows:

$$\text{Zinc, percent} = \frac{80.34A}{W}$$

where:

A = weight of zinc oxide.
W = weight of sample.

If the percentage of zinc is low a larger sample may be used, or the zinc sulfide precipitate from several samples may be combined.

Note 2.-Standard formic acid mixture shall be composed of:

Formic acid (specific gravity 1.20)	200 ml.
$(NH_4)_2SO_4$	250 gm.
Ammonia	30 ml.
Water sufficient to dilute mixture to	1,000 ml.

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Note 3.-Dilute formic acid mixture shall be composed of:

Standard formic acid mixture	25 ml.
Water sufficient to dilute to	1,000 ml.

4.4.7 Other impurities.-The sum of the percentages of volatile material, carbides, magnetic iron, alloyed iron, free metallic magnesium and zinc shall be subtracted from 100 percent. The remainder shall be percentage of other impurities.

4.5 Test applicable to type II, aluminum powder, atomized.

4.5.1 Volatile material at 105°C.-Approximately 5 gm. of the sample shall be weighed into a tared dish, heated in an oven for 3 hours at 105°C., cooled and weighed. The loss in weight shall be calculated to percentage of material volatile at 105°C., as follows:

$$\text{Material volatile at } 105^{\circ}\text{C., percent} = \frac{A \times 100}{W}$$

where:

A = loss in weight.
W = weight of sample.

4.5.2 Oil and grease.-A weighed portion of approximately 20 gm. of the sample shall be extracted with ether in a Soxhlet or similar extractor using a tared flask. The ether shall be evaporated when extraction is completed and the flask and contents shall be dried at 90°C. to constant weight, cooled in a desiccator, and weighed. A blank shall be run at the same time. The weight of residue shall be calculated to percent oil and grease, as follows:

$$\text{Percent oil and grease} = \frac{(A - B)\ 100}{W}$$

where:

A = weight of ether extract.
B = weight of blank.
W = weight of sample.

4.5.3 Zinc.-Two gm. of the sample shall be weighed into a 400 ml. beaker, covered with water and 20 ml. of concentrated HCl added slowly. When the reaction has subsided, 1 ml. of concentrated HNO_3 shall be added. The solution shall be boiled until all copper is dissolved, diluted to 200 ml., cooled to approximately 60°C., and saturated for 5 minutes with H_2S. The solution shall be filtered with a small amount of ashless paper pulp and washed with H_2S water containing a little HCl. The filtrate shall be boiled to expel the H_2S, cooled slightly and 10 gm. of citric acid added. The solution shall be diluted to 300 ml. and made neutral with ammonia, using methyl red as an indicator. Twenty-five ml. of a standard formic acid mixture (see note 4), shall be added, heated nearly to the boiling point and saturated with a rapid stream of H_2S for 15 minutes. After settling for 2 or 3 hours, the solution shall be filtered through a No. 42 Whatman or equivalent filter paper with a little paper pulp and washed with hot diluted formic acid mixture (see note 5) which has been saturated with H_2S. The zinc content shall be determined by either the gravimetric or volumetric method, using the zinc sulfide retained on the filter paper.

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Note 4.-Standard formic acid mixture shall be composed of:

Formic acid (specific gravity 1.20)	200ml.
Ammonium sulfate $(NH_4)_2SO_4$	250gm.
Ammonia	30ml.
Water sufficient to dilute mixture to	1000ml.

Note 5.-Delute formic acid mixture shall be composed of:

Standard formic acid mixture	25ml.
Water sufficient to dilute to	1000ml.

4.5.3.1 Gravimetric method.-The filter paper containing the zinc sulfide shall be ignited in a tared ignited porcelain crucible at 700°C. The residue shall be cooled in a desiccator and weighed as zinc oxide. The percent zinc shall be calculated as follows:

$$\text{Zinc, percent} = \frac{80.34A}{W}$$

where:

A = weight of zinc oxide.
W = weight of sample.

If the percentage of zinc is low a larger sample may be used, or the zinc sulfide precipitate from several samples may be combined.

4.5.3.2 Electrolytic method (see note 6).-When the percentage of zinc is lower than 0.3 percent, as is often the case for type II powder, a 3 gm. sample of aluminum shall be covered in an evaporating dish casserole, or beaker, with distilled water. The vessel shall be covered with a watch-glass and 60ml. of 20 percent (by weight) sodium hydroxide shall be added cautiously in small portions. When the reaction subsides, the solution shall be diluted to 100 ml. and boiled for 20 minutes. The solution shall be filtered while hot using a close textural filter paper, then diluted to a convenient volume and cooled to room temperature. The solution shall be electrolyzed for 30 minutes at a max. current of 1.5 amperes using a tared copper and zinc-plated cathode (see note 7), keeping the electrode completely immersed. The electrode shall be rinsed several time with distilled water, then with alcohol, dried and weighed. The increase in weight shall be calculated as the percentage of zinc in the sample. This is a rapid control method and the results obtained by it should be checked at intervals by the gravimetric or volumetric method.

Note 6.-This method is not very accurate when materially over 10 milligrams (mg.) of zinc is present in the alkaline solution of aluminum being electrolyzed.

Note 7.-The cathode shall be plated with the thinnest visible coating of copper by electrolyzing a copper sulfate solution containing a little nitric acid. On top of this a heavy zinc plate (at least 0.03 gm. zinc) shall be formed by electrolyzing a sodium hydroxide solution of zinc sulfate. The cathode is now ready for use, and may be used a large number of times without cleaning.

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4.5.3.3 Referee method for zinc.-In case of dispute, the gravimetric method shall be used.

4.5.4 Silicon as Si.-One gm. of the sample shall be weighed into a 400-ml. beaker and 30 ml. of mixed acid (see note 8) added very slowly. After solution, 5 ml. of concentrated H_2SO_4 and 12 ml. of concentrated perchloric acid ($HClO_4$) (70 to 72 percent) shall be added. The solutions shall be evaporated until copious fumes are observed, fumed 10 to 15 minutes, cooled, 150 ml. of water added, boiled and filtered through a No. 41 Whatman, or equivalent, ashless filter paper. The solution shall be washed thoroughly with hot water, dried carefully, ignited in a platinum crucible, cooled in a desiccator and weighed as silicon dioxide (SiO_2). Five ml. of hydrofluoric acid (HF) and not more than four drops of H_2SO_4 shall be added. The solution shall be evaporated, ignited, cooled in a desiccator and weighed. The loss of weight (due to the loss of SiO_2) shall be calculated to percent silicon, as follows:

$$\text{Silicon, percent} = \frac{46.72A}{W}$$

where:

A = loss in weight.
W = weight of sample.

Note 8.-The mixed acid for the silicon determination shall be composed of:

H_2SO_4 (specific gravity 1.84)	230 ml.
HCl (specific gravity 1.19)	400 ml.
HNO_3 (specific gravity 1.42)	400 ml.
Water	970 ml.

4.5.5 Free metallic aluminum (eudiometer method).-An apparatus similar to that shown on figure 2 shall be assembled. The gas measuring buret shall be graduated in 0.2 ml. The reaction flask shall contain 100 ml. of a 10 percent sodium-hydroxide solution, previously saturated with hydrogen, in the reaction flask. Means shall be provided for dropping the sample into the alkali. The weight of the sample to be used will depend on the metal content and the temperature and pressure at which hydrogen is measured. (At standard conditions of temperature and pressure a 0.33-to 0.34-gm. sample of pure aluminum will yield about 470 ml. of hydrogen.) The weight of the sample shall be roughly computed to give a volume of hydrogen within the capacity of the graduated portion of the gas buret. The system must be in equilibrium with respect to temperature and pressure and checked for any gas leakage before the determination is started. The weighed sample shall be folded in a piece of paper and placed in position. The temperature shall be recorded after comparing the reading of the thermometer in the reaction flask with a thermometer located in the close vicinity of the apparatus. The volume shall also be recorded after the pressure is equalized by means of the leveling bulb. The confining liquid in the pipet is water saturated with hydrogen gas and shall contain 0.1 percent H_2SO_4 by volume and a few drops of methyl orange to facilitate the observation of the solution in the apparatus. The sample shall be allowed to drop into the base and shall then be shaken. After the reaction has ceased, the system shall again be brought to equilibrium by cooling all parts of the system to the initial temperature, within 1°C, and again equalizing the pressures. The volume of hydrogen gas evolved shall be recorded and the percent of free metallic aluminum calculated as follows:

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$$\text{Percent free metallic aluminum} = \frac{0.0288V(P_1-P_2)}{W(273+T)} = 0.275A-1.28B$$

where:

T = temperature of water jacket in °C.
V = volume of gas in buret, ml.
P_1 = barometric pressure, in mm. of mercury.
P_2 = vapor pressure of water at T, in mm. of mercury.
W = weight of sample, gm.
A = percent zinc.
B = percent silicon

4.5.6 Alloyed iron.-An accurately weighed portion of approximately 10 gm. of the sample shall be transferred to a 500-ml. flask. About 200 ml. of 1:5 H_2SO_4 shall be added slowly to minimize effervescence and prevent spattering. The solution shall be allowed to stand until action ceases and all the aluminum has dissolved. The solution shall be titrated immediately with 0.02N $KMnO_4$ solution. The percent alloyed iron shall be calculated as follows:

$$\text{Percent alloyed iron} = \frac{5.584VN}{W}$$

where:

V = ml. $KMnO_4$ required for sample.
N = normality of $KMnO_4$ solution.
W = gm. of sample.

4.5.7 Aluminum oxide content.-The aluminum oxide content shall be determined by difference. The sum of the percentages of free metallic alluminum, iron, zinc and silicon shall be subtracted from 100 percent to determine compliance with the requirement specified.

4.6 Test applicable to type III, magnesium-aluminum alloy powdered 65/35, atomized.

4.6.1 Aluminum.

4.6.1.1 Preparation of solutions.

4.6.1.2 8-Hydroxyquinoline solution.-Five gm. of 8-hydroxyquinoline shall be dissolved in 10 ml. of glacial acetic acid. The solution shall be diluted to 100 ml. with water and filtered, if necessary.

4.6.1.3 Bromphenol blue indicator.-Into 20 ml. of 95-percent ethyl alcohol 0.100 gm. of bromphenol blue powder shall be dissolved and diluted to 100 ml. with water.

4.6.1.4 Procedure.-An accurately weighed sample of 0.7700 ± 0.0300 gm. of type III alloy shall be transferred to a 250-ml. beaker. Ten ml. of water shall be added. The beaker shall be covered with a watch glass and 20 ml. of concentrated HCl shall be added dropwise through the opening at the lip of the beaker. After the reaction has subsided, the solution shall be heated to boiling to assure complete solution. The watch glass shall be washed thoroughly and the washings drained back in to the beaker. The solution shall be filtered through a No. 41 Whatman, or equivalent, filter paper.

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The filtrate shall be caught in an accurately calibrated, 1-liter volumetric flask. The filter paper shall be washed thoroughly with hot water and the solution made up to volume with water. A 100 ml. aliquot shall be transferred to a 250-ml. beaker using an accurately calibrated buret or pipet. Three drops of bromphenol blue indicator shall be added and the solution neutralized by adding dropwise and with constant stirring, a filtered 1:1 NH_4OH solution until the yellow color turns to blue. Two drops of NH_4OH solution shall then be added in excess. The solution shall be heated to boiling and 18 ml. of the 8-hydroxyquinoline solution added in small portions, stirring after each addition to dissolve the precipitate that forms. Any precipitate remaining after the completion of the addition of the reagent shall be disregarded. The solution shall be heated to boiling and 40 ml. of 2 molar (M) ammonium acetate solution (154 gm. per liter) added dropwise with vigorous stirring. The solution shall then be placed on a steam bath for 30 minutes. The solution shall be filtered with suction through a tared, fine-porosity, fritted-glass-bottom crucible. The precipitate shall be washed eight times with 15 ml. portions of water releasing the suction before each addition and allowing the precipitate to remain in contact with the wash water for at least 1 minute prior to the application of suction. The solution shall be dried for 3 hours in an oven at 120° to 140° C., cooled in a desiccator and weighed as alluminum hydroxyquinolate, $Al(C_9H_6ON)_3$. The percent aluminum shall be calculated as follows:

$$\text{Percent aluminum} = \frac{(A - 0.0874BW)5.87}{W}$$

where:

A = gm. of precipitate.
B = percent iron (see 4.6.6)
W = gm. of sample contained in the aliquot

4.6.2 Magnesium.-A 100 ml. aliquot of the solution, prepared as specified in 4.6.1.4, shall be transferred to a 400-ml. beaker by means of an accurately calibrated buret or pipet. Three drops of bromphenol blue indicator shall be added and the solution neutralized with filtered 1:1 NH_4OH solution until the yellow color changes to blue. One gm. of ammonium chloride and 1 ml. of glacial acetic acid shall be added to the solution. Twenty ml. of hot 10-percent ammonium benzoate solution shall be added slowly while stirring. The solution shall be heated on a hot plate and boiled gently for 5 minutes. The solution shall be filtered through an 11-centimeter (cm.) Whatman No. 41, or equivalent, filter paper catching the filtrate in a 600-ml. beaker. The precipitate shall be washed 10 times with a hot wash solution containing 1 gm. of ammonium benzoate and 2 ml. of glacial acetic acid per 100 ml. of solution. Any crystallization that occurs in the filtrate as it cools shall be disregarded. The combined filtrate and washing shall be made alkaline to phenolphthalein with filtered concentrated NH_4OH solution. Five ml. of NH_4OH solution shall be added in excess. The solution shall be heated from 60° to 70°C. and 15 ml of 8-hydroxyquinoline solution added rapidly by drops from a buret. The liquid shall be stirred vigorously with a thermometer keeping the temperature of the solution at 60° to 70°C. during titration. The solution shall be heated almost to boiling with frequent stirring and transferred to a steam bath for 20 minutes. The solution shall be filtered by means of a tared, fine-porosity, sintered-glass-bottom crucible. The suction shall be disconnected and 10 ml. of warm 1:100 NH_4OH wash solution applied to the precipitate with a wash bottle agitating the precipitate as much as possible with the force of the stream of wash solution. The suction shall be applied

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until the precipitate remains only moist enough to prevent cracking. This procedure shall be repeated 7 times finally washing with two 10-ml. portions of cold water as described herein. The precipitate shall be dried to constant weight in an oven at 155° to 160°C., cooled in a desiccator, and weighed The percent magnesium shall be calculated as follows:

$$\text{Percent magnesium} = \frac{7.78A}{W}$$

where:

A=gm. of precipitate.
W=gm. of sample contained in aliquot.

4.6.3 Total aluminum and magnesium.-The percentages of aluminum and magnesium as determined in 4.6.1 and 4.6.2 shall be added together to obtain the total aluminum and magnesium.

4.6.4 Oil and grease.-A weighed portion of approximately 20 gm. of the sample shall be extracted with ether in a Soxhlet or similar extractor using a tared flask. When extraction is complete, the ether shall be evaporated, the flask and contents dried at 90°C. to constant weight. cooled in a desiccator and weighed. A blank shall be run at the same time. The weight of residue shall be calculated to percent oil and grease as follows:

$$\text{Percent oil and grease} = \frac{(A-B)\ 100}{W}$$

where:

A = weight of ether extract.
B = weight of blank.
W = weight of sample.

4.6.5 Silicon.-A weighed portion of approximately 2 gm. of the sample shall be placed in a beaker. Seventy ml. of a solution made by mixing 475 ml. of water, 125 ml. of 95 percent H_2SO_4, 200 ml. of 70 percent HNO_3, and 200 ml. of 38 percent HCl shall be added cautiously. The solution shall be evaporated until heavy fumes of sulfur trioxide (SO_3) have been evolved for a few minutes, then cooled, and 10 ml. of 50 percent H_2SO_4 added. The solution shall be diluted to 150 ml. with hot water, heated until salts are dissolved and the insoluble residue transferred to a paper filter. The residue shall be thoroughly washed with hot water and the filtrate and washings retained for use in the determination of iron (see 4.6.6). The residue shall be ignited in a platinum crucible, cooled and weighed. The residue shall be moistened with a few drops of 50-percent H_2SO_4 and 2 ml. of 47-percent HF added. The solution shall be evaporated to dryness, ignited, cooled and weighed. The loss in weight, which represents silicon dioxide, shall be calculated to percent silicon, as follows:

$$\text{Percent silicon} = \frac{46.7A}{W}$$

where:

A = loss in weight between weighings.
W = gm. of sample.

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4.6.6 Iron.-The combined filtrate and washings from the determination of silicon (see 4.6.5) shall have H_2S passed through it for 20 minutes. The precipitate shall be filtered and washed with H_2S water acidified with 1 percent H_2SO_4. The combined filtrate and washings shall be boiled vigorously for 25 to 30 minutes to remove the H_2S. The solution shall be cooled and titrated with N/10 $KMnO_4$ solution. The percent iron shall be calculated as follows:

$$\text{Percent iron} = \frac{55.8VN}{W}$$

where:

V = ml. standard $KMnO_4$ solution used in titration.
N = normality of $KMnO_4$ solution used.
W = gm. of sample taken in 4.6.5.

4.6.7 Material volatile at 105°C.-Approximately 5 gm. of the sample shall be weighed into a tared dish, heated in an oven for 3 hours and 105°C., cooled and weighed. The loss in weight shall be calculated to percentage of material volatile at 105°C, as follows:

$$\text{Material volatile at } 105^{\circ}\text{C., percent} = \frac{A \times 100}{W}$$

where:

A = loss in weight.
W = weight of sample.

4.7 Determination of average particle diameter (all types).- The average particle diameter of atomized metal powders shall be determined by means of an instrument based on the air permeability principle such as the Fisher Sub-Sieve Sizer or Picatinny Particle Sizer described below.

4.7.1 Fisher Sub-Sieve Sizer.

4.7.1.1 Apparatus.-The Fisher Sub-Siever Size (see figure 3) is composed of an air pump, and air-pressure regulator, a precision bore sample tube, a double range air flowmeter, a calculator chart for obtaining the average diameter of powders packed to a porosity greater than 0.40, and a supplementary calculator chart as supplied by Picatinny Arsenal, to be used to obtain the particle diameters of powders packed to a porosity less than 0.40, along with accessory equipment necessary to correlate these parts into a unit. The air pump builds up pressure to a constant head in the pressure regulator. The air, under this pressure head, is conducted to the packed powder sample contained in the sample tube. The flow of air through this packed bed of powder is measured by means of the calibrated flowmeter, in which the level of the water indicates directly on the charts the average diameter of the powder particles. (It should be noted that the parts of the Fisher Sub-Sieve Sizer are not replaceable with the exception of the sample tube and calculator charts. If the manometer should break or the flowmeter needs adjustment, the entire instrument must be sent to the Fisher Scientific Company for recalibration.)

4.7.1.2 Checking of apparatus.-Check the level of the water in the standpipe of the pressure regulator. The water level, as observed through the round window located in the upper left-hand side of the front panel (see figure 3), should coincide exactly with the calibration mark on the standpipe. If necessary,

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adjust the water level by adding or removing water through the glass intake arm extending through the top of the instrument cabinet. This intake arm should be stoppered except when adjustments are being made. The water level should be checked only when the sample tube has been removed from the instrument. The pilot lamp should be used to illuminate the water level. The air pump may be turned on.

Check the conditon of the drying agent in the drying tube by observing the tube through the small window in the left side of the cabinet. Use an indicating type drying agent so that the condition of the drying agent may be determined visually at any time.

Check the adjustment of the sample packing assembly as follows: Screw one of the porous plugs (see figure 4) to the plug manipulator. Lay a paper disk over one end of the sample tube and push the plug into the sample tube for a distance of approximately 0.5 inch, with the perforated surface of the plug against the surface of the paper disk, forcing the paper to crimp around the edges and precede the plug into the sample tube. Remove the plug manipulator and place the sample tube in a vertical position with the paper side of the plug up. Lay a second disk over the top of the sample tube; attach the second porous plug to the manipulator and force the plug and paper disk down into the sample tube until contact is made with the lower paper disk and plug. Remove the manipulator. Place the sample tube on the brass post beneath the rack and pinion with the lower plug in contact with the upper end of the brass post, and lower the rack until the flat bottom end comes in firm contact with the upper plug. The tip of the pointer, carried on the cross bar attached to the upper end of the rack, should coincide exactly with the base line on the calculator chart. If necessary, loosen the setscrew holding the lower brass post in its mount and adjust the height of the post until the tip of the pointer does coincide exactly with the base line.

4.7.1.3 Procedure.- Screw one of the porous plugs to the plug manipulator (see figure 4). Lay a paper disk over one end of the sample tube and push the plug into the sample tube for a distance of approximately 0.5 inch, with the perforated surface of the plug against the surface of the paper disk, forcing the paper to crimp around the edges and precede the plug into the sample tube. Place the sample tube in a vertical position with the paper side of the plug up. To make the determination on type I powder a 1.74-gram portion of dry sample is required. For type II powder a 2.70-gram portion is required. For type III powder a 2.00-gram portion is required. Transfer the weighed portion into the sample tube with the aid of a small funnel. Tap the side of the tube to settle the powder. Lay a second paper disk over the top of the sample tube; attach the second porous plug to the plug manipulator, and force the plug and paper disk approximately 2 inches down into the sample tube. Remove the manipulator. Manually press the sample contained between the filter papers to a "minimum porosity" by means of a plastic or metal rod that will fit into the sample tube. The porosity value is a measure of the degree of packing of the sample. A "minimum porosity" is reached when the sample is packed as tightly as possible by hand.

Place the sample tube on the brass post with the lower plug in contact with the upper end of the brass post. Lower the rack until the flat bottom end comes in contact with the upper plug. Turn the pinion knob as far as it will go to insure minimum porosity. Shift the calculator chart laterally until the tip of the pointer coincides with the sample height curve on the chart.

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If the sample tested has a porosity less than 0.40, use the supplementary calculator chart. (see figure 5). This chart, which is supplied by Picatinny Arsenal, can be attached to the back of the Fisher calculator chart. Adjust the supplementary chart by aligning the tip of the pointer with the base line of the chart. Make this adjustment while both plugs and both paper disks are in place in the empty sample tube. When using the supplementary chart, read the height of the sample directly in centimeters on the ordinate scale. Move the chart laterally until the tip of the pointer corresponds to the sample height line on the chart. The chart is constructed so that at this point the abscissa reading should equal the ordinate reading. Once the height of the sample is determined, the calculator chart is not moved throughout the remainder of the determination.

Adjust the initial level of the water meniscus in the manometer tube, located over the calculator chart, by means of the manometer control knob so that the meniscus coincides with the upper edge of the metal cross bar attached to the rack when the tip of the pointer coincides exactly with the base line on the calculator chart. If adjustment of the water meniscus can not be made because of too little water in the manometer tube, water may be added through the top of the manometer tube. Turn the range control knob, located at the extreme upper right of the front panel, to the "LO" position. Mount the sample tube, without disturbing the sample in any way, between the rubber-cushioned supports of the sample tube holder. Screw the upper cap down onto the sample tube until an air-tight seal is obtained at both ends. Plug the line cord into a 110 volt, 60 cycle alternating current line and throw the electrical switch at the lower right hand corner of the front panel to the "ON" position. This turns on the air pump as well as the pilot lamp which illuminates the tip of the bubbler tube in the pressure regulator standpipe as observed through the round window in the lower left-hand corner of the front panel, and the level of water in the standpipe as observed through the upper window. Adjust the pressure control knob until the bubbles rise in the standpipe at the rate of 2 to 3 per second. Allow the water level in the manometer tube to rise to a maximum level. Without disturbing the chart, turn the rack up until the upper edge of the cross bar coincides with the water meniscus in the manometer. From the location of the tip of the pointer, read the average particle diameter in microns from the chart. If the average particle diameter falls in the range of 0.2 to 20.0 microns, read the charts directly with the range control indicator to the "LO" position. If the particle size is above 20 microns turn the range control indicator to the left to the "HI" position. Multiply the chart reading by two to obtain the average particle diameter. Record the minimum porosity value.

4.7.2 Picatinny Particle Sizer.

4.7.2.1 Apparatus.-The Picatinny particle sizer is composed of an air reducing valve, an air pressure gauge, a drying tube, an air pressure regulator, a precision,bore sample tube with removable porous plug, and a multiple range flowmeter (see figures 6 and 7). Air is introduced into the apparatus through the reducing valve, at approximately 2 pounds gauge pressure. The air then

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passes through the drying bottle which is connected by means of a "T" arrangement to both the standpipe and the sample tube. The standpipe, which is kept filled with dibutylphthalate to a certain permanent level, serves to fix the pressure of the air at the point of introduction to the sample. The excess pressure is released as bubbles escaping through the dibutylphthalate. The air at this constant pressure flows downward through the sample tube in which the powder to be tested is contained. A porous plug supports the powder in the tube. After leaving the sample tube, the air is divided along two paths, a) through a porosity tube and b) into a manometer filled with dibutylphthalate. The porosity tube is made from fritted glass diffusion tubes. The manometer and the resistance formed by the porosity tube, together constitute a flowmeter. If a sample of powder is added to the sample tube the difference of pressure as read on the manometer, with and without the sample in the apparatus, will indicate the resistance of the powder sample to a constant air flow.

4.7.2.2 Calibration.

4.7.2.2.1 Checking manometer.-Check the levels of the dibutylphthalate in the manometer and the air pressure regulator before using the apparatus. If the manometer level falls below or above zero, move the scale on the manometer until the zero line corresponds with the level of the dibutylphthalate. The dibutylphthalate in the standpipe should be maintained at a constant level about five inches from the top. If additional dibutylphthalate is required, it can be added by removing the rubber stopper at the top of the standpipe and adding the liquid directly through a funnel.

4.7.2.2.2 Calibration of porosity tube.-Calibrate the porosity tube in the following manner: Assemble the apparatus with no sample in the sample tube. By means of a rubber tube, connect the porosity tube to a Sargent Wet Test Meter or equivalent. The wet test meter records the volume of air passing through it in a specified period of time. Consequently, the wet test meter, when connected to the porosity tube, measures the number of cubic feet of air passing through the porosity tube per unit time. Pass air through the wet test meter for approximately 15 minutes to determine whether it is functioning properly. Details for testing the functioning of the Sargent Wet Test Meter are explained in the "Operating Manual" accompanying the instrument. Record the volume reading as indicated by the dial on the Sargent Wet Test Meter and time reading designated by V_1 and T_1 respectively. Read the top and bottom levels of the manometer. The sum of the readings is equal to the air pressure P. After several hours, record the volume reading on the Sargent Wet Test Meter and the time of the second reading, designated V_2 and T_2, respectively. Calculate the permeability constant K of the porosity tube by means of the following formula:

$$K = 1.263 \times 10^3 \quad \frac{(V_2 - V_1)}{(T_2 - T_1)P}$$

The calibration values should be checked periodically.

4.7.2.3 Procedure.-Drop the porous plug into the sample tube with the wire gauze (325-mesh) of the plug facing up. Clamp the sample tube in a vertical position as shown on figure 6. Attach the appropriate porosity tube to the apparatus. Select the porosity tube to give a maximum precision measurement; i.e., one which will give a pressure reading of approximately one-half the pressure with the sample tube containing the porous plug only.

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Regulate the flow of the air entering the apparatus, with the air flow value, so that approximately two or three air bubbles flow through the standpipe per second. Read the top and bottom levels of the liquid in the manometer. The sum of the two readings is equal to the pressure P of the air entering the sample. Remove the sample tube and clamp it into a rubber support stand. Weigh out a sample of dry powder of unknown particle size, equal in grams to a multiple of the true density of the sample, usually five to ten times. Place the weighed sample in the sample tube with the porous plug in place and the tube clamped back into the apparatus. The air pressure is regulated as before. Tap the sample tube while air is flowing through the sample until the manometer reading is constant. Read the manometer as before. This manometer reading is equivalent to the pressure F of the air leaving the sample. Remove the sample tube and measure the height H of the sample with a calibrated measuring rod. (See figure 7.).

4.7.2.4 Calculation of air permeability particle size.-Calculate the average air permeability diameter of the particles by substituting in the following equation and solving for "mu" (μ).

$$\mu = \frac{KHN}{(AH-N)^{3/2}} \frac{F}{P-F}$$

where:

μ - the average air permeability diameter of the particles in microns. (Since it is actually a measure of surface area, it is also referred to as the average surface diameter of the particles.)

K - the permeability constant, centimeters $^{3/2}$, of the porosity tube. This constant expresses the rate of flow of air with constant head through a material of specific porosity.

H - the height of sample in centimeters.

A - the cross sectional area, square centimeters, of the tube containing the sample.

P - the initial air pressure, grams per square centimeter; i.e., the pressure of air entering the sample tube.

F - the pressure, grams per square centimeter, of the air after passing through the sample.

N - the multiplicity factor of the absolute density; i. e., the whole number which, when multiplied by the density of the material used, is equal to the weight of the sample in the sample tube.

4.8 Apparent density (all types).-A funnel having a stem of 1/4-inch internal diameter shall be cut so that the tip of the stem is just above the uppermost graduation of a tared 100 ml. graduated cylinder. The cylinder shall be filled to the 100 ml. mark by pouring small quantities of the sample on the side of the funnel and allowing it to fall by gravity through the stem. Jarring or tapping of the cylinder shall be avoided. The cylinder shall be weighed and the gain in weight divided by 100 shall be reported as the apparent density.

4.9 Granulation (all types).-A weighed portion of approximately 100 gm. of the sample shall be placed on the top sieve of the nest of sieves assembled as specified in table I, II or III, as applicable. Sieves shall comply with the requirements of Specification RR-S-366. A bottom pan shall be provided.

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The sample shall be covered and shaken for 15 minutes in a mechanical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. The portions retained by each sieve shall be weighed and calculated to a percentage basis as required to determine compliance with the granulation requirements specified in table I, II, or III, as applicable.

4.10 Resubmission and retest.-If the composite sample or any primary sample subjected to test fails to pass the tests, the lot shall be rejected. The contractor shall have the option of having a partial or complete analysis made on samples taken from all or any of the containers in the lot at no expense to the Government. The contractor may then remove the defective portions of the lot and resubmit the lot for acceptance, provided complete replacement of the defective portions can be made to the satisfaction of the inspector. The resubmitted lot shall be accepted, provided that new samples, selected in accordance with 4.2 pass all the tests required by this purchase description and that 3 additional primary samples pass any test or tests failed by the material on original submission.

5. PREPARATION FOR DELIVERY

5.1 Packing.- Unless otherwise specified on the contract or order, atomized powder shall be shipped in unlined, full open head steel drums protected against corrosion and conforming to the requirements of ICC Specification 37D stipulated in the Code of Federal Regulations 49 CFR 71-78. Containers shall contain a maximum of 150 pounds. Each drum shall be provided with a tubular rubber cover gasket, firmly cemented in place. When the jackscrew is tightened, the rim shall be tapped with a mallet to insure complete and proper sealing of the gasket. If a drum is to be reused for the shipment of atomized powder, new gaskets shall be used each time. In addition, each drum reused shall pass the tests and conditions for new drums.

5.2 Marking.- Marking to insure safe handling shall conform to the ICC Regulations stipulated in the Code of Federal Regulations 49 CFR 71-78, and in addition shipments shall be marked in accordance with Standard MIL-STD-129. Unless otherwise specified, each container shall be plainly marked with the following information completed:

(1) Minimum percentage of free metallic magnesium or aluminum.
(2) Type of atomized powder (Type I, II, or III).
(3) Nominal mesh size.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of atomized metal powder (see 1.2).
(c) Granulation and particle diameter required (see 3.1.8, 3.2.8 and 3.3.7).

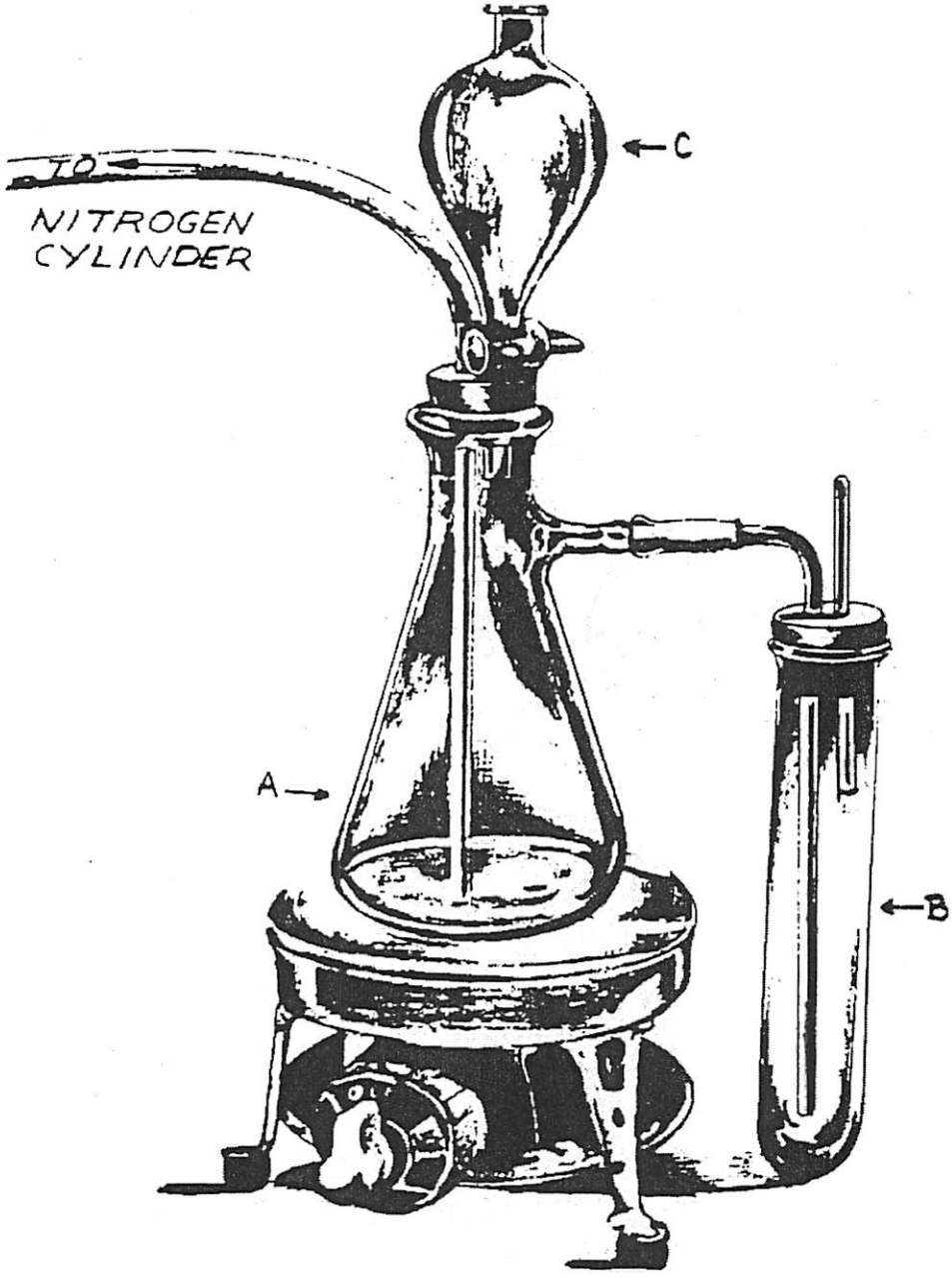
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6.2 When in the opinion of the inspector, the operation of the final processing units was not uniform during the time the material in the lot was produced, he should require a sieve analysis on each primary sample (see 4.2) or on suitable sublot composite samples.

Patent notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Army-Ordnance Corps

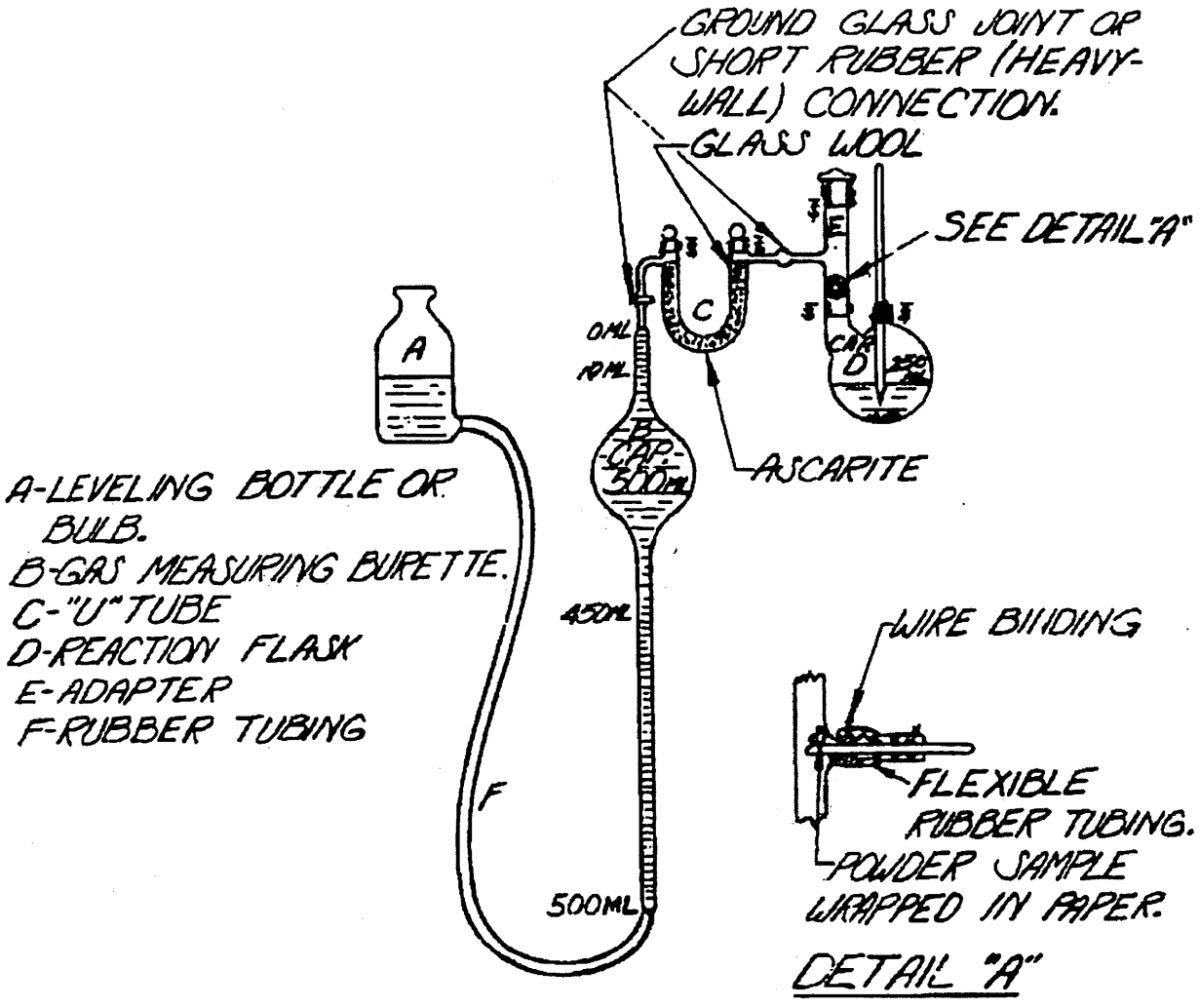
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APPARATUS FOR DETERMINATION OF CARBIDES IN MAGNESIUM

FIGURE 1

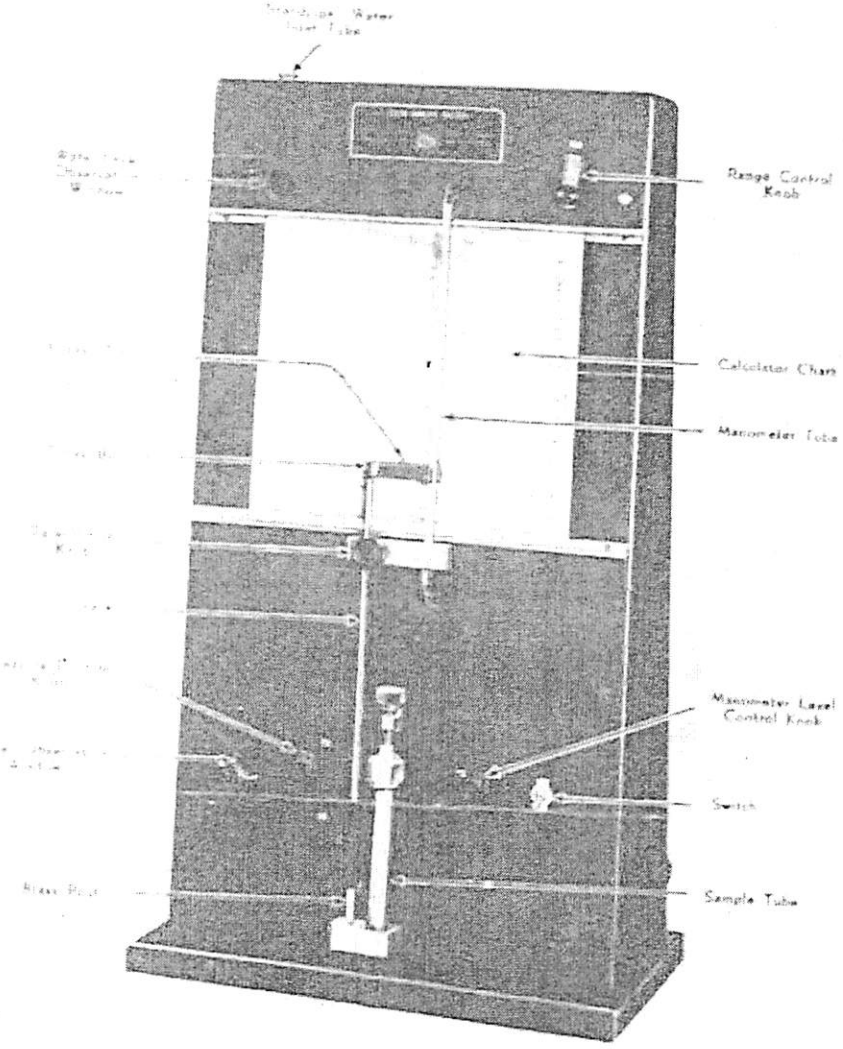
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EUDIOMETER FOR DETERMINATION OF MAGNESIUM METAL IN MAGNESIUM POWDER

FIGURE 2

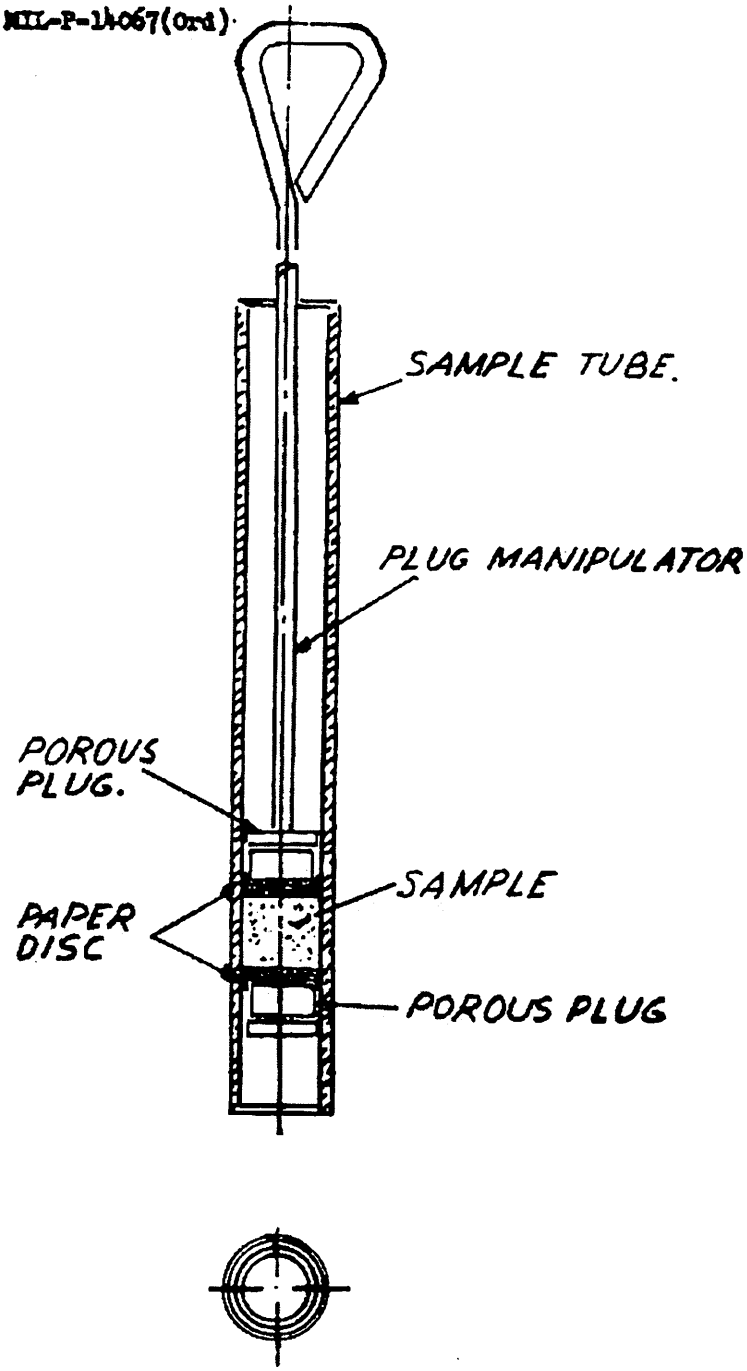
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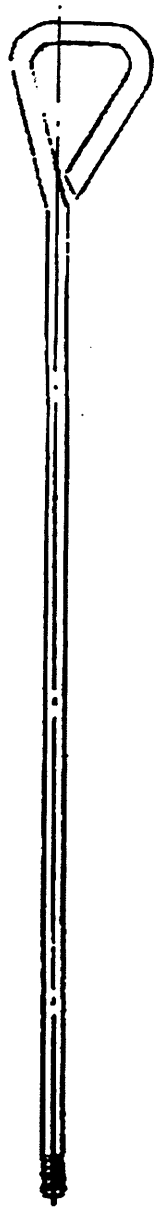
The Fisher Sub-Sieve Sizer

Fig. 3

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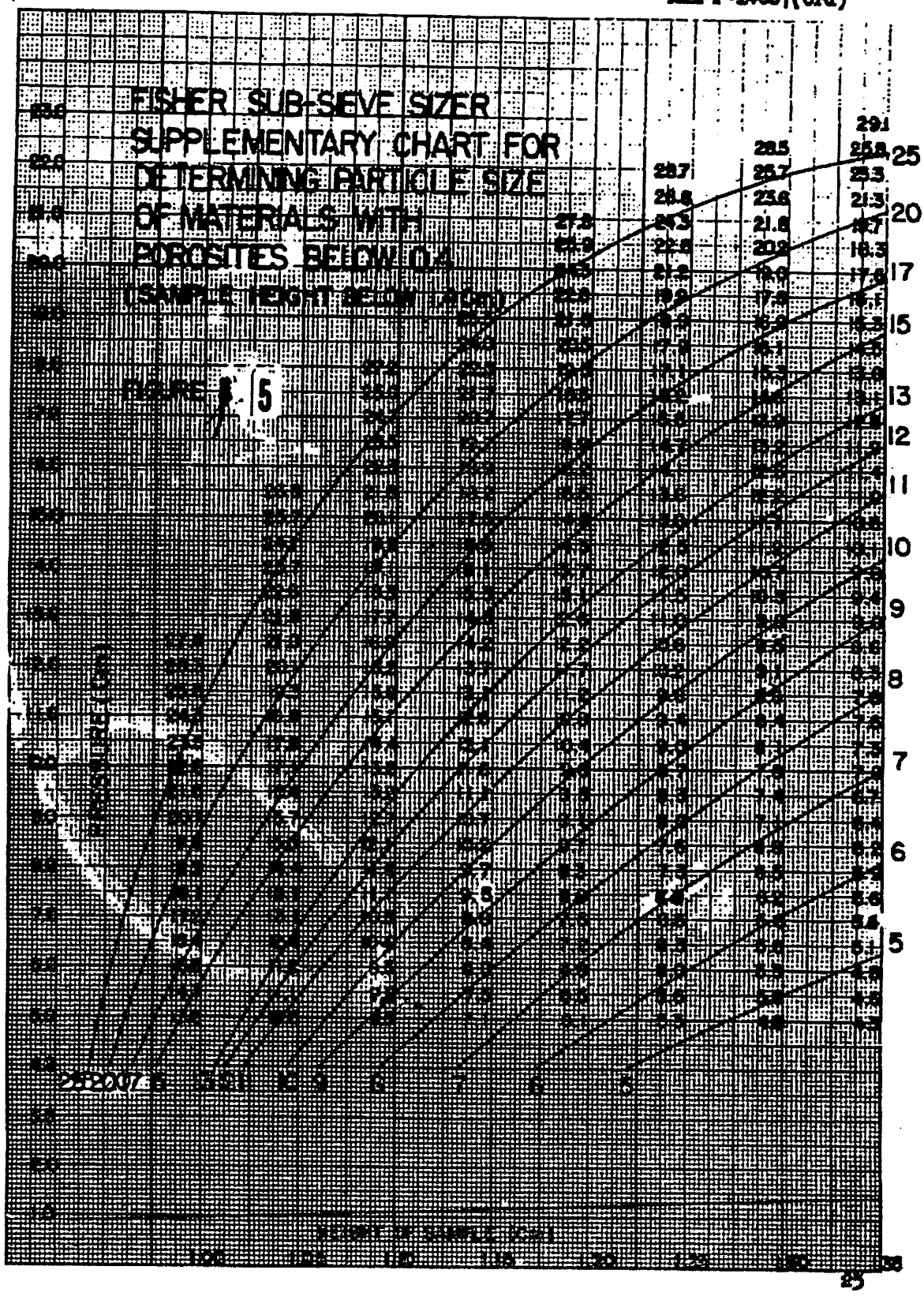
ASSEMBLY



PLUG MANIPULATOR

FIGURE 4

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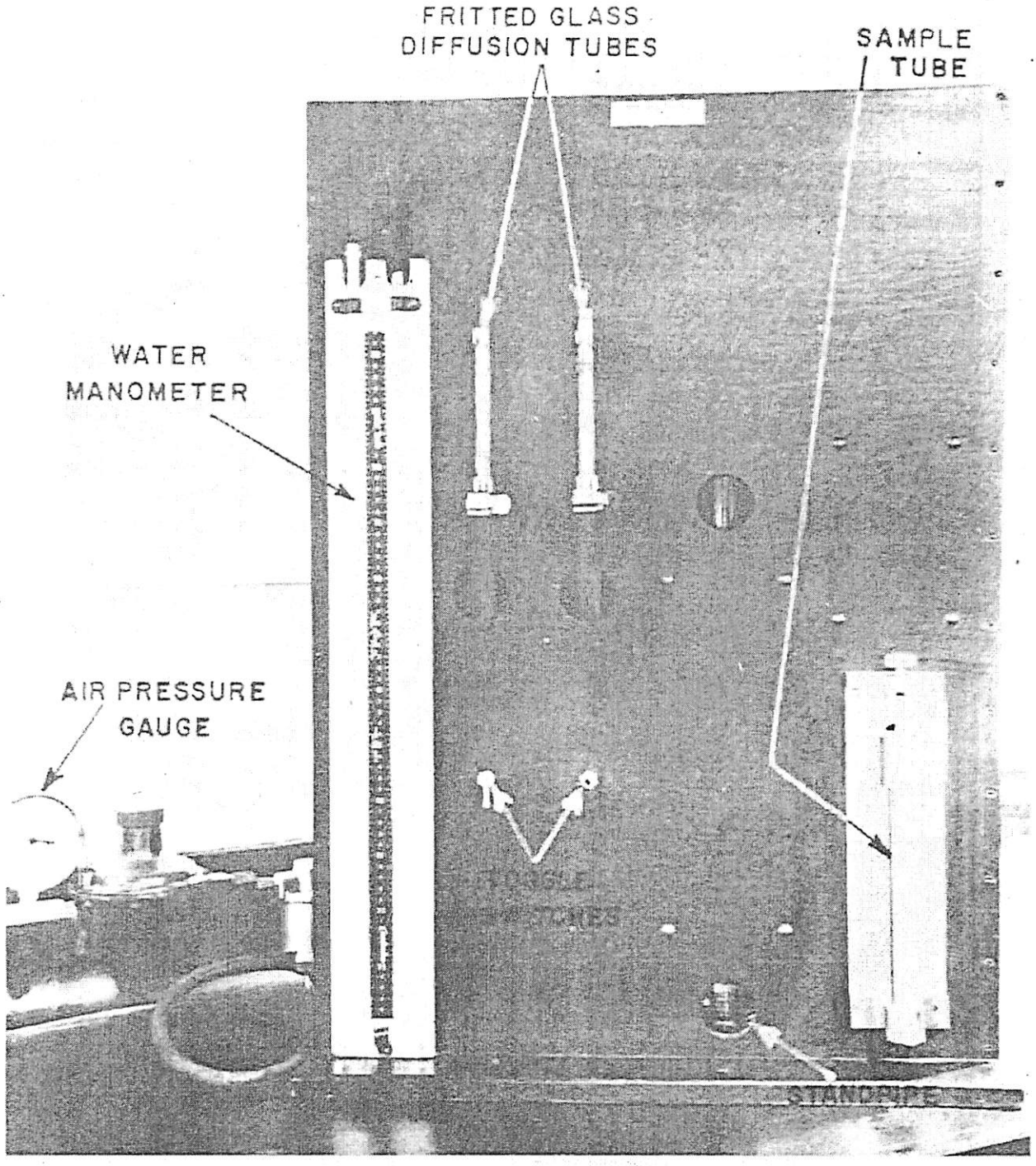


FIGURE 6A- FRONT VIEW, PICATINNY ARSENAL PARTICLE SIZER.

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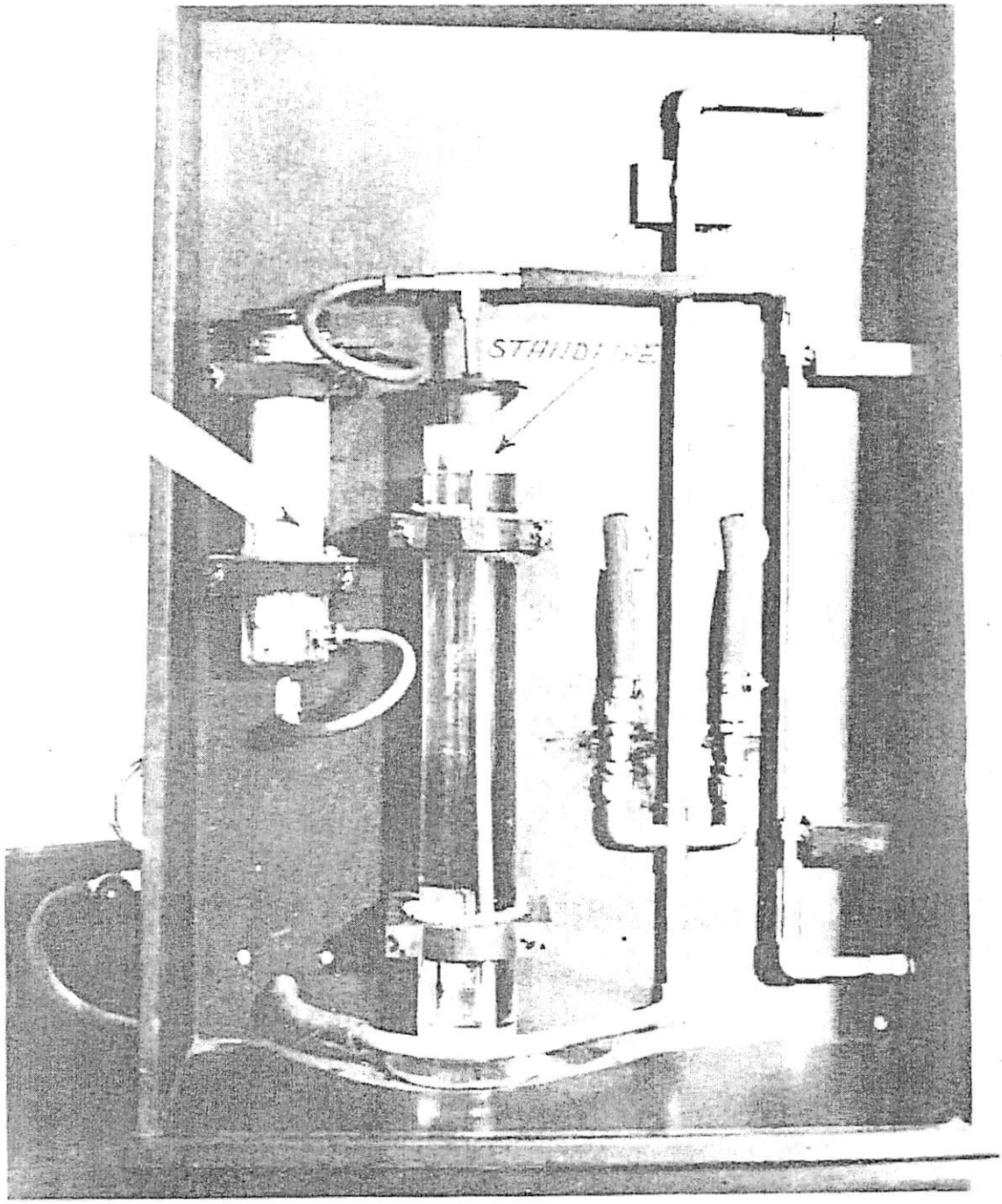
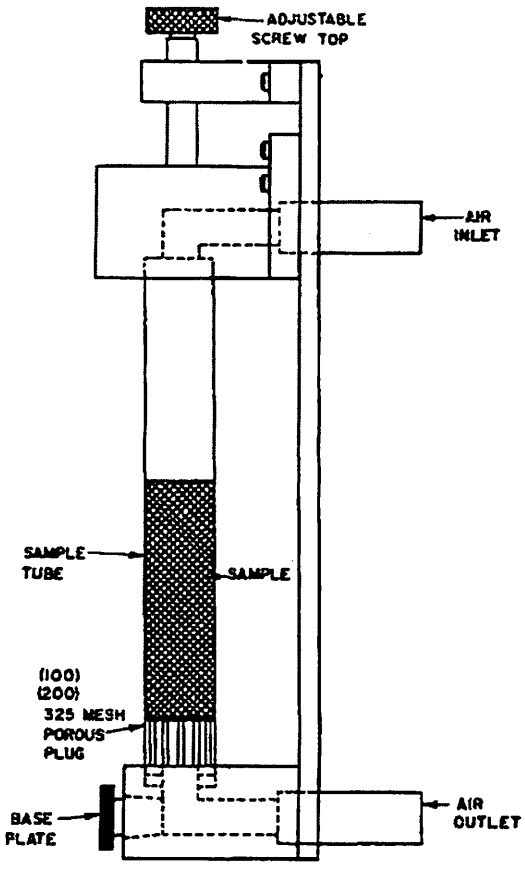
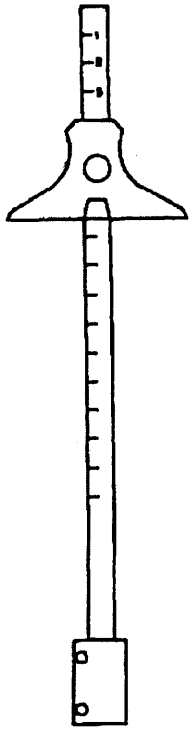


FIG. 6B BACK VIEW, PICATINNY ARSENAL PARTICLE SIZER.

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SAMPLE TUBE ASSEMBLY



SAMPLE HEIGHT GAUGE

FIGURE 1 - SAMPLE TUBE ASSEMBLY AND SAMPLE HEIGHT GAUGE, PICATINNY ARSENAL PARTICLE SIZER.

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